

Table IV.3: Statistical Measures per Scenario

Scenario	Iterations	Workload Sizes	Primary Metric	Statistical Approach
Scenario 1: Key Establishment	2,000 per algorithm	Single operation (no size variable)	Execution time (ms)	Median, IQR, 5th–95th percentile, skewness
Scenario 2: Hash Functions	2,000 per input size	1 KB and 1 MB	Latency (ms) and Throughput (MB/s)	Median, IQR, 5th–95th percentile, skewness
Scenario 3: AES Encryption	200 per file size	1 KB, 10 MB, 100 MB	Encryption time (ms) Throughput (MB/s)	Mean, std dev, 95% CI, skewness, Shapiro-Wilk
Scenario 4: Hybrid Cryptography	2,000 per message size	1 KB and 1 MB	End-to-end latency (ms) Component split	Median, IQR, 5th–95th percentile, skewness